$\Delta H = nC_p (T_f - T_i)$

$\Delta H = 2 \times \frac{5}{2} R (298 - 398)$

$\Delta H = - 500 R$

17.

$C + O_2 \rightarrow CO_2$ $\quad \Delta H^0 = - a$ KJ ..... (i)

$2CO + O_2\ 2CO_2$ $\quad \Delta H^0 = -b$ kJ ..... (ii)

$C + \frac{1}{2} O_2 \rightarrow CO\ \Delta H^0 = ?$

(i) × 2

$2C + 2O_2 \rightarrow 2CO_2$ $\quad \Delta H^0 = - 2a$ kJ ..... (iii)

Reverse of equation (ii) will be

$2CO_2 \rightarrow 2CO + O_2$ $\quad \Delta H^0 = + b$ kJ ..... (iv)

(iii) + (iv)

$2C + O_2 \rightarrow 2CO\ \Delta H^0 = b - 2a$ kJ ..... (v)

(v) ÷ 2

$C + \frac{1}{2} O_2 \rightarrow CO\ \ \Delta H^0 = \frac{(b - 2a)}{2}$ kJ

18.

Given :

$\Delta H_C (CH_4) = - 890$ kJ mol^{-1}

$\Delta H_C (C_3H_8) = - 2220$ kJ mol^{-1}

Let the mixture contain lit of and lit of propane.

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

x $\quad$ 2x

$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

(15.68 – x) $\quad$ 5 (15.68 – x)

Volume of oxygen consumed = 2x + 5 (15.68 – x) = 32 lit

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